

PRODUCT REQUIREMENTS DOCUMENT

Blocking Chart Creator

MediaOS Module

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Initiative (IPG Mediabrands) | Unilever Canada
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1. Problem Statement

1.1 Who Is Affected

Media planners and managers at Initiative who build blocking charts for Unilever campaign activations. This includes the planner (Danielle Cheung), the manager (Lucus Dato), and any assistant media planners supporting the process. Across the Foods & Wellbeing portfolio alone, this team builds 15 to 20 blocking charts per year across Hellmann's, Knorr, Liquid IV, and OLLY.

1.2 Current Workflow

The blocking chart is the single source of truth for a campaign's media plan. It captures every tactic, platform, audience, budget, and fee calculation in a standardized Excel template that feeds downstream into traffic sheets (via QuickClick), Prisma campaign setup, and client-facing presentations.

Today, creating a blocking chart requires a planner to manually translate data from two separate KSO activation response workbooks (one for Social, one for Programmatic) into the blocking chart template. Each activation response uses a different column layout, different terminology, and different structural conventions. The planner must:

1. Open both activation response workbooks and identify the correct tabs (Social: Media Plan tab; Programmatic: KSO Media Plan_TTD+ADSP and KSO Media Plan_DV360 tabs)
2. Mentally map each column from the activation response to the corresponding blocking chart column (these differ between Social and Programmatic)
3. Determine which section each tactic belongs to (Programmatic Video vs. Connected TV vs. Social) based on platform and media type
4. Translate terminology (e.g., 'Programmatic Guaranteed (PG) via Investment Team' becomes 'Managed Service'; 'Channel' column values become 'Brand Say Social', 'Other Say Social', etc.)
5. Split combined audience targets into primary and secondary rows (the activation response puts 'M/F 18-54 3P Flavour Seekers + Sports Enthusiast - 1PD CRM + LAL' in one cell; the blocking chart needs this across two rows)
6. Apply the correct fee formula set based on channel type (DV360 gets Scibids 7%; Social gets different DV Postbid rates; Linear/OOH has zero fees)
7. Apply planner-level strategic adjustments (reallocating budgets between platforms, removing or adding tactics, splitting budgets by EN/FR ratio)
8. Set up all SUM formulas for section headers, total rows, and header area references
9. Merge cells for platform groups, measurement blocks, and secondary target rows
10. Highlight cells where information is missing or TBD so stakeholders know what still needs resolution

1.3 Pain Point

This process takes 2 to 4 hours per blocking chart depending on campaign complexity. It is entirely manual, error-prone, and requires deep institutional knowledge of fee structures, terminology translations, and formatting conventions. Errors in fee calculations or section assignments are caught late (often during Prisma reconciliation), causing rework.

1.4 Impact

Metric	Value
Time per chart	2 to 4 hours (average 3 hours)
Charts per year	15 to 20 across Foods & Wellbeing portfolio
Annual time cost	45 to 60 hours of planner/manager time
Error rate	Fee miscalculations found in ~30% of charts during Prisma reconciliation
Rework cost	1 to 2 additional hours per chart when errors are caught
People affected	2 to 3 planners per account; scales to all Initiative teams using the same template

1.5 Evidence

- Validated through direct observation: Hellmann's Soccer/BBQ 2026 blocking chart built from two activation responses as a prototype exercise, documenting every transformation rule and decision point
- QuickClick (blocking chart to traffic sheet automation) already saves 2+ hours per traffic sheet, proving downstream automation works and users adopt it
- Performance review flagged 80+ hour weeks driven partly by execution work like this that should be automated or delegated

2. Product Vision

Planners spend their time on strategic decisions, not data translation between spreadsheets.

The Blocking Chart Creator transforms KSO activation responses into audit-ready blocking charts in minutes. The planner's role shifts from manual data entry to strategic review: adjusting budgets, approving tactics, and making the decisions that only a human can make.

3. Proposed Solution

3.1 What It Does

The Blocking Chart Creator is a module within MediaOS that accepts activation response files as input, applies a configurable set of transformation rules, and outputs a fully formatted blocking chart in the standardized Initiative template.

3.2 User Experience

Before (Current State)

Planner opens two activation response Excel files, opens a blank blocking chart template, and spends 2 to 4 hours manually copying, translating, and formatting data. They reference institutional knowledge for fee formulas, section assignments, and terminology mapping. Errors are discovered days later during Prisma setup.

After (With Blocking Chart Creator)

Planner logs into MediaOS, navigates to the Blocking Chart Creator module within VSMFA, and uploads activation response file(s) via drag-and-drop. The system parses the data in seconds, then presents an interactive adjustment panel where the planner can reallocate budgets between platforms (with sliders), exclude or add tactics, set EN/FR splits, and define BLS structure. A live preview of the blocking chart updates as adjustments are made. Once satisfied, the planner exports a completed blocking chart Excel file with all formulas, formatting, merges, and yellow highlights for missing data. The output can feed directly into QuickClick for traffic sheet generation.

3.3 Core Transformation Engine

The engine performs these operations in sequence:

1. **Parse:** Read activation response Excel file(s), extract tactic data from specified tabs
2. **Map:** Apply column-level transformations (see Section 4 for complete mapping rules)
3. **Adjust:** Apply planner configuration (budget reallocations, platform exclusions, EN/FR splits, BLS structure)
4. **Populate:** Write transformed data into the blocking chart template, preserving all formatting, formulas, and structural conventions
5. **Validate:** Run checks (budget reconciliation, formula integrity, section coverage, yellow highlighting for missing data)

4. Transformation Rules

4.1 Source File Structure

Social Activation Response

Primary tab: Media Plan. Columns A through O covering Measurement, Flight Dates, Platform, Objective, Campaign Details, Channel (Brand Say / Other Say), Ad Formats, KPI, Target, Budget, Est. Reach, Est. Impressions, Est. CPM, Est. Frequency, KPI Volume. EN and FR sections are separated within the tab with a percentage split defined at the top.

Programmatic Activation Response

Two tabs: KSO Media Plan_TTD+ADSP and KSO Media Plan_DV360. Columns B through O covering Platform, Objective, Campaign Details, Target, Buy Type, Language/Geo, KPI, KPI Benchmark, Est. Viewability, Start Date, End Date, Net Budget, Est. Impressions, Est. CPM. EN and FR campaigns are separated into labeled sections.

4.2 Column Mapping

Blocking Chart Col	Source Field	Transformation Rule
B: Channel	Social col F	Map 'Brand Say' / 'Other Say' + append 'Social'. Append ' - Cell A' or ' - Cell B' for BLS multi-cell rows only.
C: Platform	Social col C / Prog col B	Populate only on first row of each platform group within a section. Values: TikTok, Meta, Pinterest, YouTube, TradeDesk, Amazon, Twitch.
D: Media Type	Derived	Social = 'Social'; YouTube/Twitch = 'Digital Video'; CTV platforms = 'Connected TV'; Linear = 'Linear'; OOH = 'Experiential Out of Home'. First row of platform group only.
E: Buy Type	Social col D / Prog col F	Translate: 'PG via Investment Team' = 'Managed Service'; DV360 'CPM' = 'Auction'; 'Reach & Frequency' / 'Reach (Reserved)' = 'Reach & Frequency'.
F: Objective	Social col D / Prog col C	Direct map: Awareness, Reach (Auction), Reach (Reserved), Traffic, Video Views.
G: Placements	Social col E / Prog col D	Copy directly, preserving line breaks for multi-line descriptions.
K: Language	Section position / Prog col G	'EN' or 'FR' derived from section placement or Language/Geo field.
P: Target	Social col I / Prog col E	CRITICAL: Split at ' - ' separator. Primary audience in main row, each secondary audience gets its own row with only P populated.

Q: Est. CPM	Social col M / Prog col O	Dollar value, direct transfer.
R: Impressions	Formula	Always calculated: $=U[\text{row}]/Q[\text{row}]*1000$. Never taken from activation response.
U: Gross Budget	Social col J / Prog col M	Primary input. All fees and net budget derive from this value.
V-AB: Fees	Formula	Calculated per fee rules (see Section 4.4).

4.3 Section Assignment

Platform / Placement Type	Blocking Chart Section
Linear TV (Specialty, Conventional, Integrations)	LINEAR TELEVISION
Experiential activations (hot dog stands, sky writing)	OUT OF HOME
YouTube, Twitch, Online Video	PROGRAMMATIC VIDEO
Amazon Prime Video, Netflix, Sportsnet, Fubo, TVA+, Radio Canada	CONNECTED TV
TikTok, Meta, Pinterest, Reddit, Snapchat	SOCIAL

Key distinction: content served on a TV screen (streaming apps, CTV) goes under Connected TV. Online video (YouTube, Twitch) goes under Programmatic Video.

4.4 Fee Calculation Rules

Channel Type	V: Ad Serving	W: Scibids	Y: DV Prebid	Z: DV Postbid	AA: Buffer
Linear / OOH	0	0	0	0	0
DV360 / YouTube	$\text{Imps}/1000 * 0.0169$	$\text{Net} * 0.07$	$\text{Imps}/1000 * 0.027$	$\text{Imps}/1000 * 0.2$	$\text{Fees} * 1.3$
TTD / Amazon	$\text{Imps}/1000 * 0.0169$	0	$\text{Imps}/1000 * 0.027$	$\text{Imps}/1000 * 0.2$	$\text{Fees} * 1.3$
Social (all)	0	0	$\text{Imps}/1000 * 0.027$	$\text{Imps}/1000 * 0.1013$	$\text{Fees} * 1.3$

Net Budget (AB) is always: $=U[\text{row}] - AA[\text{row}]$. Data Fees (X) default to 0.

4.5 Planner Adjustment Configuration

The planner's strategic decisions are captured as a configuration layer that sits between raw activation data and the final output. This is the 'planner brain' that differentiates a thoughtful media plan from a mechanical data transfer.

Config Parameter	Description
budget_reallocations	Move budget from one platform to another (e.g., Reddit \$40K to Pinterest, split 80/20 EN/FR)

platform_exclusions	Remove platforms entirely (e.g., exclude Fubo)
en_fr_split	Default EN/FR percentage split for platforms without pre-defined language budgets (e.g., 80/20)
bls_structure	Define BLS eligibility per platform/language (multi_cell, single_cell, not_applicable, not_eligible)
cpm_overrides	Override activation response CPMs with historical performance data
added_tactics	Tactics not in activation response (e.g., reserved buys like Pulse Premiere, TopView)

5. Validation and Output

5.1 Automated Checks

- Budget reconciliation: Total Gross Budget (F11) matches expected total from activation responses after adjustments
- EN/FR split verification: Language budget ratios match expected percentages
- Formula integrity: Every section header SUM covers all its data rows; every total row references its section header; grand total SUMs all channel totals
- Secondary target rows: Verified to have no values outside column P
- Fee rule compliance: Scibids 7% only on DV360/YouTube rows; DV Postbid rates correct per channel type
- Zero formula errors after recalculation

5.2 Yellow Highlighting

Any cell where information is missing or marked TBD gets a yellow highlight (RGB: FFFF00) so stakeholders immediately see what needs resolution. This applies to: Viewability (M) on digital rows, KPI Value (O) when TBD or empty, Start/End Dates (S/T) when empty.

5.3 Output Format

The output is the standard Initiative blocking chart Excel template (.xlsx) with all formatting, merges, formulas, column widths, and number formats preserved exactly as the current template. This is non-negotiable for audit compliance and account standardization.

6. Build Plan

6.1 Phase 1: MVP (4 to 6 weeks)

Scope: Web-based module within the MediaOS platform (Next.js/React) that accepts activation response file(s) via drag-and-drop upload, presents an interactive planner adjustment panel, and exports a completed blocking chart.

- Supports Unilever account template only
- Handles Social (TikTok, Meta, Pinterest) and Programmatic (DV360, TTD, Amazon) activation responses
- All fee calculation rules hardcoded per Section 4.4
- Planner adjustments via interactive UI panel (budget sliders, platform toggles, EN/FR split controls, BLS structure selector)
- Validation checks from Section 5.1 run automatically

Success metric: Blocking chart creation time drops from 2 to 4 hours to under 15 minutes (including planner review and adjustment).

6.2 Phase 2: MediaOS Integration (6 to 8 weeks after Phase 1)

- Web-based UI within MediaOS with drag-and-drop file upload
- Advanced planner adjustment panel with historical CPM suggestions and campaign-level budget optimization
- Preview mode showing the blocking chart before export
- Template selector for multi-client support (Unilever, Merck)
- Direct pipeline to QuickClick (blocking chart output feeds traffic sheet generation)

6.3 Phase 3: End-to-End Automation (Future)

- DAB form parsing: Extract brief data to pre-populate activation requests
- Blocking chart to Prisma import automation
- Fee rule engine with configurable rates (accounts for rate changes without code updates)
- Historical CPM benchmarking from past campaign data

6.4 What Is NOT Included in Phase 1

- No multi-client template support (Unilever only in Phase 1)
- No advanced analytics or optimization recommendations
- No Prisma integration
- No DAB form parsing
- No real-time platform data pulls (works from exported Excel files only)

7. Technical Approach

7.1 Technology Stack

- Next.js/React frontend (consistent with existing MediaOS/QuickClick architecture), hosted within MediaOS sandbox infrastructure
- Node.js/Python backend for Excel parsing (pandas/openpyxl), transformation engine, and formula recalculation (LibreOffice)
- User authentication and analytics via MediaOS sandbox infrastructure (compliance-approved, web-based deployment)

7.2 Data Flow

User uploads Activation Response Excel(s) via MediaOS UI --> Parse Engine extracts tactic data --> Planner Adjustment Panel presents configuration options --> Transform Engine applies mapping rules + adjustments --> Populate Engine writes to template --> Validate Engine runs checks --> Preview displayed in-browser --> Export .xlsx on approval. Output can optionally feed directly into QuickClick for traffic sheet generation.

7.3 Risks and Mitigations

Risk	Impact	Mitigation
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KSO changes activation response format	Parser breaks	Schema validation on input; alert if columns don't match expected layout
Fee rates change (Scibids, DV)	Incorrect calculations	Externalize rates in config file; Phase 3 adds rate management UI
New platforms added to activation	Unmapped data	Default 'unknown' section with yellow highlight; flag for planner review
Template changes (new columns, shifted rows)	Output breaks	Template version tracking; validation against expected column headers

8. Success Metrics

Metric	Target	Measurement
Time savings	85% reduction (3 hours to under 30 minutes including review)	Self-reported time tracking before/after
Error rate	Zero fee calculation errors in output	Prisma reconciliation audit (post-campaign)
Adoption	100% of Unilever blocking charts use the tool within 3 months	Usage logs
Template compliance	Output passes audit review with zero formatting deviations	Finance/audit team spot checks

8.1 Failure Criteria

- Planners revert to manual process because output requires more than 15 minutes of corrections
- Fee calculations are less accurate than manual process
- Template formatting deviates from standard (audit risk)

9. Resource Ask

Resource	Commitment	Timeline
Product/Dev (Lucus)	Product management, front-end development, user research (Lucus)	Weeks 1 through 6
Engineering	1 backend developer for transformation engine, API, Excel generation	Weeks 2 through 6
QA / Testing	Danielle + 1 planner from another team for validation	Weeks 4 through 6

Decision needed by: March 20, 2026 to begin development and target Q2 delivery.

Appendix A: Prototype Artifact

A complete working prototype was built for the Hellmann's Soccer/BBQ 2026 campaign, translating two KSO activation responses (Social and Programmatic) into a fully formatted blocking chart. This prototype validated every transformation rule documented in this PRD and produced a blocking chart with:

- 47 data rows across 5 sections (Linear TV, OOH, Programmatic Video, Connected TV, Social)
- 265 formulas with zero errors after recalculation
- Budget reconciliation to \$1,917,500 gross (\$1,867,351 net)
- Strategic adjustments applied: Reddit \$40K reallocated to Pinterest (80/20 EN/FR), Fubo \$40K moved to Amazon Prime Video
- BLS multi-cell structure preserved (Cell A / Cell B labeling on Meta EN rows)
- Yellow highlighting on 11 cells with missing data
- Measurement column merged across platform/tactic blocks

The prototype Excel file and detailed mapping specification (Markdown) are available as companion artifacts to this PRD.